

Experiment - 4.1.1. Set Operations

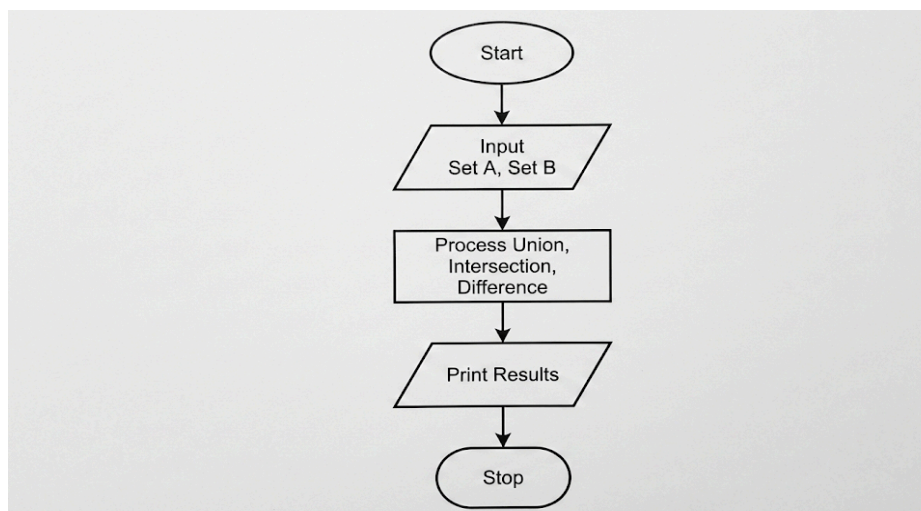
1. Aim

To design and implement a Python program that performs mathematical set operations, specifically union, intersection, and difference, on two sets: Set A and Set B. The program prompts the user to input space-separated integers for both sets and computes the results.

2. Pseudocode

1. **START**
2. **PRINT** the prompt "Set A: " and **READ** a line of space-separated integers.
3. **CONVERT** the input values into a mathematical set and **STORE** it in the variable A.
4. **PRINT** the prompt "Set B: " and **READ** a line of space-separated integers.
5. **CONVERT** the input values into a mathematical set and **STORE** it in the variable B.
6. **CALCULATE** the union of Set A and Set B ($A \cup B$) and **STORE** the result in union_set.
7. **CALCULATE** the intersection of Set A and Set B ($A \cap B$) and **STORE** the result in intersection_set.
8. **CALCULATE** the difference between Set A and Set B ($A - B$) and **STORE** the result in difference_set.
9. **PRINT** the text "Union:" followed by the value of union_set.
10. **PRINT** the text "Intersection:" followed by the value of intersection_set.
11. **PRINT** the text "Difference:" followed by the value of difference_set.
12. **END**

3. Flowchart



4. Python Program

Read sets

```
A = set(map(int, input("Set A: ").split()))
```

```
B = set(map(int, input("Set B: ").split()))
```

Operations

```
union_set = A | B
```

```
intersection_set = A & B
```

```
difference_set = A - B
```

Output

```
print("Union:", union_set)
```

```
print("Intersection:", intersection_set)
```

```
print("Difference:", difference_set)
```

5. Experiment Screenshot

The screenshot displays the C++DE TANTRA IDE interface. On the left, the problem description for "4.1.1. Set Operations" is visible, detailing the input format (two lines of space-separated integers for Set A and Set B) and the output format (three lines showing Union, Intersection, and Difference). Below the description are sample test cases. The main editor area on the right contains the Python code implementing the set operations. The code reads two sets, A and B, and calculates their union, intersection, and difference. The execution results at the bottom show that the program passed 2 out of 2 shown test cases and 2 out of 2 hidden test cases. The expected output for the first test case is displayed, showing the Union, Intersection, and Difference for the given sets.

4.1.1. Set Operations

Write a Python program to perform union, intersection and difference operations on *Set A* and *Set B*.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for *Set A*
- The second input prompts "Set B: " followed by space-separated list of integers for *Set B*

Output Format:

- The first line prints "Union: " followed by the union of *Set A* and *Set B*
- The second line prints "Intersection: " followed by the intersection of *Set A* and *Set B*
- The third line prints "Difference: " followed by the difference of *Set A* and *Set B*

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases

Test case 1

Set A: 0 2 4 5 8

Set B: 1 2 3 4 5

Union: {0, 1, 2, 3, 4, 5, 8}

Intersection: {2, 4, 5}

Difference: {0, 8}

Test case 2

Set A: 10 11 22 33 44 55

Set B: 15 16 17 18 19 20

Union: {10, 11, 15, 16, 17, 18, 19, 20, 22, 33, 44, 55}

Intersection: set()

Difference: {10, 11, 22, 33, 44, 55}

setoperat...

```
1 # Read sets
2 A = set(map(int, input("Set A: ").split()))
3 B = set(map(int, input("Set B: ").split()))
4
5 # Operations
6 union_set = A | B
7 intersection_set = A & B
8 difference_set = A - B
9
10 # Output
11 print("Union:", union_set)
12 print("Intersection:", intersection_set)
13 print("Difference:", difference_set)
14
15
```

Average time: 0.008 s
Maximum time: 0.009 s
7.75 ms
9.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 4ms

Expected output

Set A: 0 2 4 5 8

Set B: 1 2 3 4 5

Union: {0, 1, 2, 3, 4, 5, 8}

Intersection: {2, 4, 5}

Difference: {0, 8}

Actual output

Terminal Test cases

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